

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Align probe for max signal / perpendicular to field (at known distance) (1) Put probe at a fixed/known distance (1) Current calculated from equation $B = \frac{\mu_0 I}{2\pi a}$ (1)	3			3	1	1
	(b)		Use of $F = BIl$ (1) Use of $B = \frac{\mu_0 I}{2\pi a}$ (1) $\left[\text{or } \frac{F}{l} = \frac{\mu_0 I^2}{2\pi a} \text{ for 2 marks} \right]$ Answer = $0.336[4 \text{ N m}^{-1}]$ (1) Repulsive force (or equivalent) (1)	1 1	1 1		4	3	
	(c)	(i)	Use of Bqv [i.e. $F = 2.15 \times 10^{-14} \text{ N}$] (1) $\frac{mv^2}{r}$ or $m\omega^2 r$ used (1) Radius calculated $r = 3.56 \text{ c[m]}$ or $F = 3.06 \times 10^{-14} \text{ [N]}$ or $v = 2 \times 10^7 \text{ [m s}^{-1}\text{]}$ (1) Not circular or radius too large or field not uniform or force not large enough or speed too large (1)			4	4	2	
		(ii)	Starts in correct direction (1) Curving downward or clockwise (1)		2		2		
	(d)		Equation $F = Eq$ used (1) $F = Bqv$ used (1) Note $Eq = Bqv$ or $E = Bv$ scores both marks Substitution of $E = \frac{V}{d}$ (1) (use of $V = Bvd$ gets 3 marks) Final correct calculation(s) e.g. forces = $(2.15 \text{ \& } 2.16) \times 10^{-14} \text{ N}$ or confirmation of V , B , v or d from the other values (1) Final conclusion consistent with answer ecf - true (accept not true because 2.15 different from 2.16) (1)			5	5	3	
			Question 4 total	5	4	9	18	9	1